

Yash Agarwal

AI/ML Engineer · M.Tech, IIT Jammu

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Education

Degree	Institute	CGPA	Year
M.Tech (CSE)	Indian Institute of Technology, Jammu	7.34	2024–Present

Thesis

•Measuring Robustness of Pruned Models

2025-26

Under Prof. Dr. Yamuna Prasad Shukla, HOD, IIT Jammu

- Built an end-to-end prune→train→attack pipeline benchmarking **5 PaI methods** (SNIP, GraSP, SynFlow, DPaI, YOPO) across **VGG19/ResNet18/ResNet20**, 7 sparsity levels, and 5 attacks (FGSM, PGD, AutoAttack).
- Designed the **RTI and normalized nRTI metrics** to isolate mask-topology robustness from source-data effects on CIFAR-10→CIFAR-100 transfer.
- Demonstrated that **data-independent mask computation predicts cross-dataset robustness transfer** ($r \approx 0.97$).
- Identified **data-free methods** (YOPO, SynFlow) as the strongest transferable subnetworks, ranking top-2 across all architectures.
- Established that adversarial robustness is partly **encoded in mask topology at initialization**, with the transfer gap scaling with data-dependence.

Projects

•Intelligent Account Servicing Workflow (IASW)

2026

React, FastAPI, LangChain, OpenAI, Google Document AI, MySQL, ChromaDB

- Built a **Human-in-the-Loop (HITL) banking automation system** that processes customer account-change requests while ensuring AI never writes to the core banking system autonomously.
- Designed a **5-stage AI pipeline** (validation, document storage, OCR, confidence scoring, summarization), staging each request until a human checker approves.
- Integrated **Google Document AI OCR** with a **GPT-4o vision fallback** and **LangChain** prompt chains generating structured checker-facing summaries.
- Engineered **duplicate-document detection** using **ChromaDB** SHA-256 fingerprints and a **difflib confidence scorer** (PASS/FLAG/FAIL at 85%/60% thresholds) producing audit-ready score cards.

•Financial Document Question Answering System (RAG-based)

2026

Python, LangChain, OpenAI, ChromaDB

- Built an **LLM-powered RAG system** answering complex financial queries over SEC 10-K filings (Amazon, Uber), cutting manual document-review time significantly.
- Generated **OpenAI embeddings** stored in **ChromaDB** via **RecursiveCharacterTextSplitter** chunking for scalable top-K semantic retrieval.
- Designed an end-to-end **LangChain pipeline** (RunnablePassthrough → ChatOpenAI → StrOutputParser) returning grounded, citation-backed answers.

•Automated Image Caption Generator

2025

Python, TensorFlow, DenseNet201, LSTM

- Built an **encoder-decoder captioning model** pairing **DenseNet201** feature extraction with an **LSTM decoder**, trained on **Flickr8k** (8,000 images).
- Achieved a **BLEU-1 score of 63.4%** (~25% over baseline) via optimized tokenization, vocabulary pruning, and padded-sequence handling.

•Quora Question Pair Similarity

2025

Python, Scikit-learn, XGBoost

- Trained **Random Forest** and **XGBoost** classifiers on **400K+ Quora question pairs** using TF-IDF, token-overlap, and fuzzy-match features for duplicate detection.
- Achieved **~80% accuracy**, outperforming a TF-IDF cosine-similarity baseline by **~7%**.

Technical Skills

Languages: Python, C++, Java, SQL **ML / Deep Learning:** PyTorch, TensorFlow, Scikit-learn, XGBoost, Computer Vision, NLP

GenAI & LLMs: RAG, LangChain, Hugging Face, Vector Databases (ChromaDB), OpenAI API

Automation & Tools: n8n Workflow Automation, Postman, Power BI, Flask

Databases: MongoDB, MySQL, PostgreSQL, ChromaDB **Cloud:** Microsoft Azure

Achievements

- **GATE 2024 (CS)** – Scored **676**; ranked top **1%** among 125,000+ candidates nationwide.
- **LeetCode Knight** – Earned the Knight badge for a contest rating in the top **~5%** of competitive programmers on LeetCode.
- **Amazon Summer School 2025** – Selected among top **3,000 students** across India; completed an ML curriculum led by Amazon scientists.